BRAC UNIVERSITY

Principles of Physics-II (PHY-112)

Department of Mathematics and Natural Sciences

Quiz: 03, **Section**: 02

Date: May 5, 2025

Duration: 30 Minutes | **Spring 2025 (10F-29C)** | Marks: 15

Use SI Units only. Partial Marks will be given for partially correct answers ONLY.

1. You see an unknown charged particle (observer) move toward a potential downhill, when released from rest, in the presence of an external field from a line charge. The observer charge carries — **(1)**

   ◯ positive charge ✓ ◯ negative charge ◯ no charge at all ◯ Need more information to comment

   (Alternative) You see an unknown charged particle (observer) move toward a potential uphill, when released from rest, in the presence of an external field from a surface charge. The observer charge carries — **(1)**

   ◯ positive charge ◯ negative charge ✓ ◯ no charge at all ◯ Need more information to comment

2. A planar-shaped equipotential surface would be a suitable Gaussian surface for a— **(1)**

   ◯ line charge ◯ point charge ◯ volume charge ◯ plane charge ✓

   (Alternative) A cylindrical-shaped equipotential surface would be a suitable Gaussian surface for a— **(1)**

   ◯ line charge ✓ ◯ point charge ◯ volume charge ◯ plane charge

3. Which statement claims that the Electric Field is always perpendicular to the potential surfaces **(1)**

   ◯ $\oint \vec{E} \cdot d\vec{l} = 0$ ✓ ◯ $\epsilon_0 \oint \vec{E} \cdot d\vec{a} = Q_{\text{enc}}$ ◯ $\vec{\nabla} \cdot \vec{E} = 0$ ◯ $\oint \vec{E} \cdot d\vec{a} = 0$

4. Crossing two points of the same equipotential surface ($\Delta V = 0$ on the surface) will give you— **(1)**

   ◯ always a constant $\vec{E}$ field ◯ no $\vec{E}$ field at all ✓ ◯ $\vec{E}$ field has nothing to do with potential surfaces
   ◯ $\vec{E}$ field opposite to your path of motion

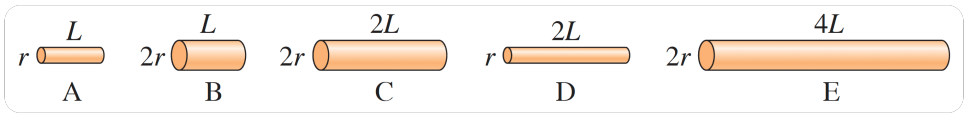
5. What are (i) the charge and (ii) the charge density on the surface of a conducting sphere of radius 2 cm whose Potential is 200 V (with $V = 0$ at infinity)? **(3)**

   Class 9 — Incepting Ideas #2

   (Alternative) An empty hollow metal sphere has a potential of -500 V with respect to ground (defined to be at $V = 0$) and has a charge of -5.0 nC. Find the electric Potential at the center of the sphere. Explain your answer. **(3)**

   Class 9 — Incepting Ideas #3

6. The wires are all made of the same material. A fixed potential ΔV is applied across all wires. Rank in order, from smallest to largest, the resistances R_A to R_B or currents I_A to I_E. **(2)**



   Class 11 — Incepting Ideas #5, Q1

7. A very long insulating cylinder of charge of radius 2.50 cm carries a uniform linear density of 15 nC m^{-1}. If you place one probe of a voltmeter at the surface, how far from the surface must the other probe be placed so that the voltmeter reads 100 V? **(3)**

   Class 9 — Incepting Ideas #2, Q2

8. An electric field of strength E is inside a parallel-plate capacitor of d spacing. An electron is released from rest at the negative plate. What is the electron's speed when it reaches the positive plate? **(3)**

$$\Delta V = Ed$$

$$\Delta KE = \frac{1}{2}mv^2$$

$$\Delta KE = q\Delta V$$

$$\frac{1}{2}mv^2 = qEd$$

$$E = \frac{mv^2}{2qd}$$